



Indian Institute of Technology Madras



Driving e-Mobility

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Department of Engineering Design

Centre of Excellence for Zero Emission Trucking

About CoEZET

- ⚡ Established at Department of Engineering Design.
- ⚡ Objective: Accelerating adoption of Zero Emission Trucks (ZET) in India.
- ⚡ Will become a Nodal Resource for India ZET Solutions.

e-Mobility Focus

- 🏠 Research in Electric Vehicles (EV) domain – Automotive control, Batteries, Charging, Power Electronics, Electric Motors, Controllers & Vehicle integration, Vehicle Dynamics and Control
- 🏠 Academic Programs & Courses on eMobility
- 🏠 Centre of Excellence for Zero Emission Trucking (CoEZET)

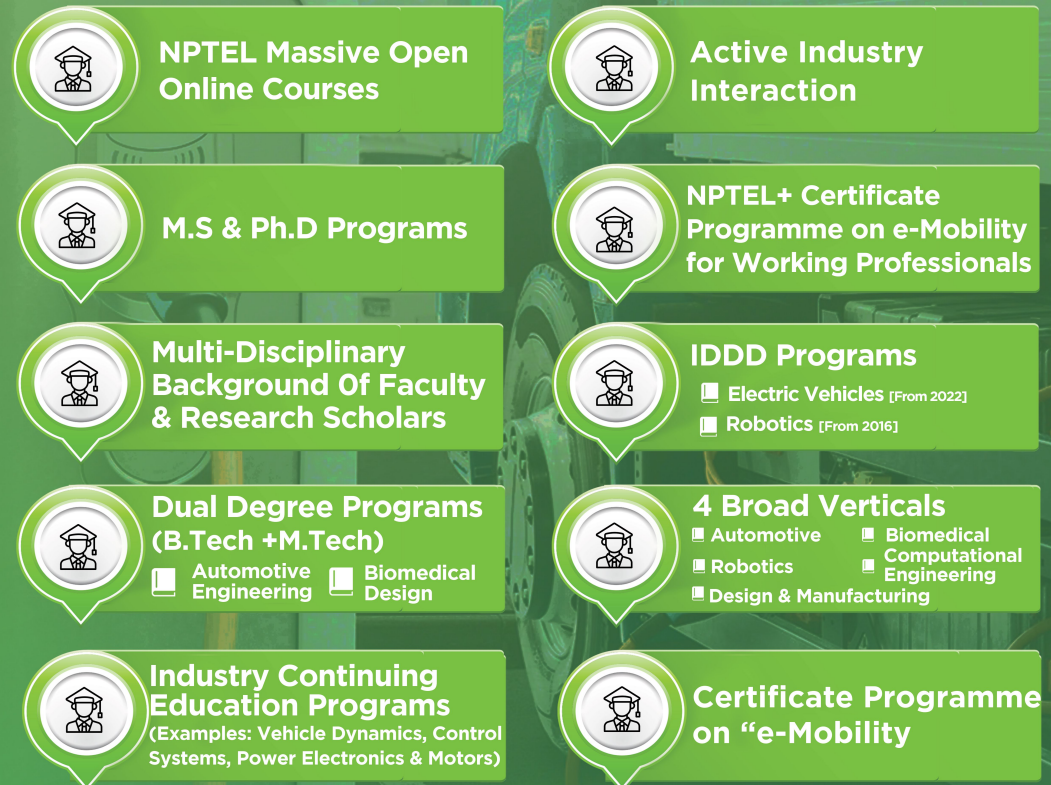
Department of Engineering Design & e-Mobility

Innovating since 2006

Pioneering Indigenous Engineering Solutions – Automotive | Biomedical | Robotics | Computational Engineering | Design & Manufacturing



ACADEMICS



Target Audience:

Automotive Engineers (in Original Equipment Manufacturers, Component Suppliers, Service Providers, Aftermarket Dealerships), & for those intending to upskill themselves in the eMobility domain.

Centre for Outreach and Digital Education (CODE), IIT Madras
<https://code.iitm.ac.in/cpoem-cohort-4>

COLLABORATE WITH US

- 🔌 Funded Research Projects
<https://icandsr.iitm.ac.in/icandsr/>
- 🔌 Retainer Consultancy: Short-term engagement, specific requirements
- 🔌 Research-Based Industry Projects: Long-term engagement, significant research focus
- 🔌 PhD and MS Programs: External Mode,
<https://research.iitm.ac.in/>
- 🔌 Continuing Education Programs:
<https://code.iitm.ac.in/>

PROJECTS FOR CSR FUNDING

S.NO	PROJECT TITLE	CONTACT PERSON (NAME & EMAIL)	DURATION (IN MONTHS)	BUDGET (IN LAKHS)	DESCRIPTION
1	Passive Component Sizing for High-Voltage Power Electronic Systems in Hybrid Electric Vehicles (HEVs)	Prof. Srikanthan Sridharan (srikanthan@iitm.ac.in)	24	30	The project aims to systematically determine the appropriate size of passive components in the traction path power electronic converters in HEVs using a comprehensive multi-physics approach including thermal, ripple containment, switching transient, resonance and stability considerations.
2	Design, development, & control of rare-earth magnet-free traction motors	Prof. Srikanthan Sridharan (srikanthan@iitm.ac.in)	36	60	The project aims to design, and build an emerging group of low-cost, high speed and torque-dense configurations of rare-earth magnet-free motors (both PM-assisted and non-assisted) for traction applications and develop control strategies to maximize the dynamic efficiency of the drive system.
3	Design & Development of Battery Charger for Universal Electric Vehicle Charging Applications	Prof. Deepak Ronanki (dronanki@iitm.ac.in)	36	100	The project aims to design and develop a universal charging infrastructure that is able to charge the battery packs with a wide output voltage range (120 V to 900 V) with a higher efficiency and better power transfer capability.
4	Design & Development of High-Power Wireless Battery Charging Systems for Electric Trucks	Prof. Deepak Ronanki (dronanki@iitm.ac.in)	36	135	The project aims to design and develop high-power wireless battery charging systems with size of 400 mm X 400 mm for high-battery voltage systems (700 V) suited for electric trucks.
5	Fuel Cell Hybrid Powered Highly Efficient Power Conversion E-Mobility Systems	Prof. Deepak Ronanki (dronanki@iitm.ac.in)	48	105	The project aims to perform studies with different combinations of energy sources under different drive patterns and to develop highly efficient integrated power conversion systems for fuel-cell-hybrid-based electric vehicles.
6	Intelligent Fast Charging Systems with High Battery Longevity for e-Mobility Applications	Prof. Deepak Ronanki (dronanki@iitm.ac.in)	36	125	The project aims to design and development of an intelligent battery charger to reduce the charging time without affecting the life cycle of the battery regardless of battery chemistries.
7	Design & Development of Fault-Tolerant Electric Powertrains for Electrified Vehicles	Prof. Deepak Ronanki (dronanki@iitm.ac.in)	24	60	The project aims to design and develop fault-tolerant power conversion systems for electrified vehicles under fault conditions to operate limp-home conditions.
8	Development of advanced machine learning techniques for dynamic state of the charge estimation & temperature control-based charging techniques	Prof. Jayaganthan (edjay@iitm.ac.in, metarj@gmail.com)	36	80	The project aims to develop dynamic state-of-charge estimation methods and charging techniques using advanced artificial intelligence-based methods.
9	Development of bidirectional battery charger for G2V and V2X applications	Prof. Deepak Ronanki (dronanki@iitm.ac.in)	36	60	The project aims to develop a bidirectional battery charger that facilitates the power flow from grid to vehicle, vehicle to grid, vehicle to vehicle and vehicle to X.

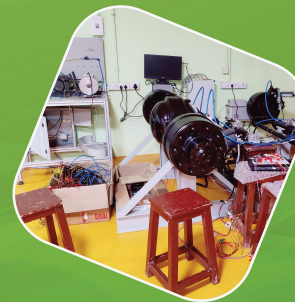
FACILITIES



**Charging
Laboratory**



**Virtual Proving
Ground**



**Automotive Control
Laboratory**



**Battery Engineering
Laboratory**

**Power Electronics &
e-Drive Laboratory**



**Automotive
Engineering
Laboratory**

Facilities for CSR Funding *(Upcoming Labs)*

S. NO	LAB NAME	BUDGET (IN LAKHS)
1	Battery Engineering Laboratory	135
2	Battery Charging Laboratory	83
3	Electric Vehicle Integration Laboratory	116
4	Motor and Controller Laboratory	120
5	Power Electronics Laboratory	73